

haps worth mentioning that all these codes can be viewed as particular punctured subcodes of the extended Golay code $(24, 2^{12}, 8)$, cf. [10]. This permits an easy determination of their automorphism groups, as was done by Berlekamp [11] for the Nordstrom–Robinson codes.

REFERENCES

- [1] M. Nadler, "A 32-point $n = 12, d = 5$ code," *IRE Trans. Inform. Theory*, vol. IT-8, p. 58, 1962.
- [2] J. H. van Lint, "A new description of the Nadler code," *IEEE Trans. Inform. Theory*, vol. IT-18, pp. 825–826, 1972.
- [3] R. F. Stevens and W. G. Bouricius, "The heuristic generation of large error correcting code," unpublished research memorandum, IBM Res. Center, Yorktown Heights, NY, August 1959.
- [4] F. J. MacWilliams, A. M. Odlyzko, and N. J. A. Sloane, "Bounds for codes of length less than 25," to be published.
- [5] M. W. Green, "Two heuristic techniques for block code construction," *IEEE Trans. Inform. Theory*, vol. IT-12, p. 273, 1966.
- [6] A. W. Nordstrom and J. P. Robinson, "An optimum nonlinear code," *Information and Control*, vol. 11, pp. 613–616, 1967.
- [7] P. Delsarte, "Bounds for unrestricted codes by linear programming," *Philips Res. Repts.*, vol. 27, pp. 272–289, 1972.
- [8] —, "Four fundamental parameters of a code and their combinatorial significance," *Information and Control*, vol. 23, pp. 407–438, 1973.
- [9] S. L. Snover, "The uniqueness of the Nordstrom–Robinson codes and the Golay binary codes," Ph.D. dissertation, Michigan State University, Dept. of Math., 1973.
- [10] J. M. Goethals "On the Golay perfect binary code," *J. Combinatorial Theory*, vol. 11, pp. 178–186, 1971.
- [11] E. R. Berlekamp, "Coding theory and the Mathieu groups," *Information and Control*, vol. 18, pp. 40–64, 1971.

On Fast M -Sequence Transforms

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Abstract—The equivalence of M -sequence matrices to Walsh–Hadamard matrices can be exploited to take advantage of the latter's fast transform algorithm. The nature of the obvious equivalence, its relation to the M -sequence and its implementation as address modifications realized by linear feedback shift-registers are discussed.

I. INTRODUCTION

An M -sequence is a binary sequence generated by a linear feedback shift-register whose characteristic polynomial is primitive. An M -sequence can be shown to have an impulse-like autocorrelation function [1]; for this reason M -sequences are often called "pseudo-noise sequences," and their distinct cyclic permutations, or phases, form a useful signaling alphabet [1]–[3]. A drawback is that correlation-detection at the receiver can entail on the order of p^2 operations, where p , the period of the sequence, is of the form $2^n - 1$, and where n is the degree of the feedback equation. Clearly, such computation can grow unwieldy even for modest sized n . The cost of this computation can be drastically reduced by exploiting the equivalence between the M -sequence matrix and the Walsh–Hadamard matrix. The latter has a "fast-transform" cost of $p \cdot \log_2 p$ additions for a matrix multiplication, a feature that has been successfully used in the rapid decoding of first-order Reed–Muller codes [3]–[5].

While the equivalence between M -sequence matrices and Walsh–Hadamard matrices has been recognized for some time [6], [7], our purpose here is to specify the equivalence in a manner

conducive to implementation. To this end, we prove that the equivalence is permutational on both sides, and relate the indicated address modifications to two shift-registers naturally associated with the sequence. The argument consists of demonstrating factorizations of binary M -sequence matrices and binary Walsh–Hadamard matrices. From these factorizations we infer row and column permutations which relate the two kinds of matrices, and we characterize the permutations in terms of the M -sequence. Because the equivalence is permutational, it holds under any renaming of matrix entries; in particular, it holds for matrices with ± 1 entries, which embody the real-variable M -sequence and Walsh–Hadamard transforms. Finally, it should be noted that the transform described here is just one of a class, whose structure is thoroughly discussed in [7].

II. M -SEQUENCE MATRICES

An M -sequence $\{s_k\}$ is a binary sequence (not all zeroes) generated by a primitive linear recursion over $GF(2)$, the finite field of integers modulo 2. A primitive recursion of degree n

$$s_{k+n} = \sum_{j=0}^{n-1} c_j s_{k+j} \quad (1)$$

generates an M -sequence of period $p = 2^n - 1$, so that $s_{k+p} = s_k$, for all k . To each primitive recursion of the form (1), there corresponds a primitive polynomial

$$f(x) = x^n + \sum_{j=0}^{n-1} c_j x^j \quad (2)$$

which is called the characteristic polynomial of the sequence $\{s_k\}$. In the sequel, we shall use the notation

$$[s_k] = (s_0, s_1, \dots, s_{p-1}) \quad (3)$$

for a single period of $\{s_k\}$ beginning with the n -tuple $s_0 = s_1 = \dots = s_{n-2} = 0$ and $s_{n-1} = 1$.

The M -sequence matrix $M = (m_{ij})$, associated with an M -sequence $\{s_k\}$, is a square matrix of order p with

$$m_{ij} = s_{i+j-2(\text{modulo } p)}, \quad i, j = 1, 2, \dots, p. \quad (4)$$

Thus, the first row of M is $[s_k]$ and the successive rows are simply successive cyclic shifts of $[s_k]$. Equation (4) also makes clear that the matrix M is symmetric.

As an example for $n = 3$, consider the recursion $s_{k+3} = s_{k+1} + s_k$. The characteristic polynomial for the sequence generated by this recursion is $f(x) = x^3 + x + 1$ and $\{s_k\} = (0010111)$. Hence, the corresponding matrix M is given by

$$M = \begin{bmatrix} 0 & 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}. \quad (5)$$

From the construction of M , it is clear that each row and each column of M satisfy the recursion (1) which generates $\{s_k\}$. Therefore, every row (column) of M can be expressed as a linear, modulo-two combination of the first n rows (columns) of M and we can write

$$M = LS = S'L' \quad (6)$$

where L is a binary matrix of order $p \times n$, where S is the $n \times p$ matrix formed by the first n rows of M , and where the prime denotes matrix transposition.

Since all the rows of M are distinct, all the rows of L must be distinct, therefore, every nonzero binary n -vector must appear as some row in L . Also, the first n rows of L form an identity matrix of order n .

Now, let σ be the square matrix of order n , which is formed by the first n columns of S . From (6) we have

$$L\sigma = S'. \quad (7)$$

Since every nonzero binary n -vector appears as a row in both L and S' , the matrix σ is necessarily nonsingular, and we get

$$L = S'\sigma^{-1}. \quad (8)$$

In our example for $n = 3$, we have

$$S = \begin{bmatrix} 0 & 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 & 1 & 0 & 0 \end{bmatrix} \quad \sigma = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \quad \sigma^{-1} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

and

$$L = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

The reader may easily verify that the product LS will result in the matrix M given by (5).

A further investigation of the matrix $L = (l_{ij})$ reveals that for all $i = 1, 2, \dots, p$, the polynomial

$$g_i(x) = \sum_{j=1}^n l_{ij}x^{j-1} \quad (9)$$

is congruent to x^{i-1} modulo the characteristic polynomial $f(x)$ of the M -sequence $\{s_k\}$. That is,

$$g_i(x) = x^{i-1} \pmod{f(x)}, \quad i = 1, 2, \dots, p. \quad (10)$$

To prove relation (10), we first observe that it holds trivially for $i = 1, 2, \dots, n$, because the first n rows of L form an identity matrix of order n . Now we proceed by induction on i . Suppose (10) holds for all $i < k$, $k \geq n$, and consider the power x^k of x . To find the residue of x^k modulo $f(x)$, we multiply both sides of (2) by x^{k-n} to get

$$x^{k-n}f(x) = x^k + \sum_{j=0}^{n-1} c_j x^{j+k-n}. \quad (11)$$

Since the left side of (11) is zero modulo $f(x)$, we have

$$x^k = \sum_{j=0}^{n-1} c_j x^{j+k-n} \pmod{f(x)}. \quad (12)$$

By the induction hypothesis, (12) may be rewritten as

$$x^k = \sum_{j=0}^{n-1} c_j g_{j+k-n+1}(x) \pmod{f(x)},$$

which, together with (9), gives

$$\left. \begin{aligned} x^k &= \sum_{j=0}^{n-1} \sum_{r=1}^n c_j l_{j+k-n+1,r} x^{r-1} \\ &= \sum_{r=1}^n x^{r-1} \sum_{j=0}^{n-1} c_j l_{j+k-n+1,r} \end{aligned} \right\} \pmod{f(x)}. \quad (13)$$

Now, by (8), the columns of L are linear combinations of the columns of S' . Therefore, each column of L satisfies the recursion (1), which generates $\{s_k\}$; hence,

$$\sum_{j=0}^{n-1} c_j l_{j+k-n+1,r} = l_{k+1,r}. \quad (14)$$

Substituting (14) into (13), we get

$$x^k = \sum_{r=1}^n l_{k+1,r} x^{r-1} = g_{k+1}(x) \pmod{f(x)}.$$

This completes the proof of relation (10).

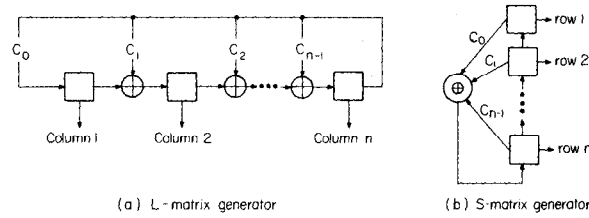


Fig. 1. Feedback generators for the matrices L and S .

In our example, $f(x) = x^3 + x + 1$, so that direct computation results in

$$\left. \begin{aligned} x^0 &= 1 \\ x^1 &= x \\ x^2 &= x^2 \\ x^3 &= 1 + x \\ x^4 &= x + x^2 \\ x^5 &= 1 + x + x^2 \\ x^6 &= 1 + x^2 \end{aligned} \right\} \pmod{f(x)}.$$

The reader may easily verify that the same results are obtained by applying (9) and (10).

The foregoing discussion suggests simple circuit structures to generate the factors L and S of M . The successive rows of L are obtained as successive states of the feedback circuit shown in Fig. 1(a), while successive columns of S are obtained as successive states of the ordinary feedback-shift-register shown in Fig. 1(b). In both circuits the values of the gating variables c_i are those of the recursion (1), so that the circuit of Fig. 1(b) is just the generator of the sequence $\{s_k\}$.

As a consequence of the factorization $M = LS$, multiplication by M over $GF(2)$ entails a cost proportional to $p \log_2 p$, consisting of multiplication first by S (n inner products of dimension p) and then by L (p inner products of dimension n). Unfortunately, the factorization is invalid over the reals, and further discussion is required to prove that the real M -sequence transform can also be realized at this reduced cost.

III. HADAMARD MATRICES

A fast transform method currently in use has an organizational structure exemplified by the matrix below.

$$\begin{array}{c} \text{indices} \end{array} \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 & 1 & 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 1 & 1 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}$$

This structure occurs in coding theory as a Reed-Muller matrix or as the McDonald matrix [4], [6], [7]; in transform theory it characterizes the Walsh (or Walsh-Hadamard) transform [8]. In the latter context, the transforms are defined over the real values ± 1 rather than over $GF(2)$, the correspondence being

$$\begin{array}{ccc} GF(2) & & \text{Reals} \\ 0 & \Leftrightarrow & +1 \\ 1 & \Leftrightarrow & -1. \end{array} \quad (15)$$

The structure of H for a given n can be described in several

ways.

- i) Construct (over the reals) the n th Kronecker power of

$$\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix};$$

then replace each 1 by 0 and each -1 by 1.

- ii) Construct the modulo-two Kronecker sum of

$$\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

with itself n times.

- iii) The columns of H can be viewed as the columns of a standard n -variable truth-table and their modulo-two vector sums in all 2^n possible combinations. Denoting the basis columns by n unit vectors, whose binary-numerical equivalents are $1, 2, 4, \dots, 2^{n-1}$, one can assign all the columns indices in the natural way and arrange them accordingly.

This last description shows that like M in the previous section, H can be factored as [9]

$$H = B \cdot B' \quad (16)$$

where

$$B = \begin{bmatrix} 0 \\ 1 \\ 2 \\ \vdots \\ 2^{n-1} \end{bmatrix},$$

and where k denotes the binary representation of the number k .

For example, when $n = 3$, we have

$$B \cdot B' = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 & 0 & 1 \end{bmatrix} = H.$$

The factorizations (6) of M and (16) of H suggest the relationship between the two matrix types and a means of converting one to the other.

IV. THE EQUIVALENCE OF $\hat{M}$ AND H

A. Permutation Equivalence

Let $\hat{L}$ and $\hat{S}$ be the matrices obtained from L and S by bordering the former on top with a row of all zeros and the latter on the left by a column of all zeros. The bordered matrix $\hat{M}$ is then given by

$$\hat{M} = \hat{L}\hat{S}. \quad (17)$$

The factors $\hat{L}, \hat{S}$ of $\hat{M}$, and B, B' of H have in common the fact that each exhausts, either as its rows or as its columns, all the possible binary n -vectors. In B , they occur in natural binary order; in $\hat{L}$ and $\hat{S}$, their order is determined by the natural state-sequence of the appropriate linear feedback device. But since the same vectors occur, $\hat{L}$ and B differ only by a row permutation, while $\hat{S}$ and B' differ only by a column permutation. Therefore, we can write

$$\hat{L} = P_L B \quad \hat{S} = B' P_S \quad (18)$$

where P_L and P_S are $2^n \times 2^n$ permutation matrices describing the respective n -vector orderings in $\hat{L}$ and $\hat{S}$. (Things are this simple because the rows of B and columns of B' appear in natural numerical order.)

From (16), (17) and (18), it is immediate that

$$\hat{M} = \hat{L}\hat{S} = P_L B B' P_S = P_L H P_S. \quad (19)$$

Moreover, since this equivalence is permutational, (19) holds even when the 0,1 entries of M and H are replaced by ± 1 according to (15). Therefore, an M -sequence transform can be accomplished by

- i) reordering the received data according to P_S ;
- ii) performing a Walsh-Hadamard transform; and
- iii) reordering the transformed data according to P_L .

In this manner the economy of a fast-transform algorithm may be realized.

REFERENCES

- [1] S. W. Golomb, *Shift Register Sequences*. San Francisco: Holden-Day, 1967.
- [2] —, *Digital Communications with Space Applications*. Englewood Cliffs, NJ: Prentice-Hall, 1964.
- [3] G. D. Forney, "Coding and its application in space communications," *IEEE Spectrum*, vol. 7, pp. 47-58, June 1970.
- [4] R. R. Green, "A serial orthogonal decoder," *Jet Propulsion Laboratory Space Program Summary* 37-39, vol. 4, pp. 247-252, 1966.
- [5] E. C. Posner, "Combinational structures in planetary reconnaissance," in *Error Correcting Codes*, (Ed. H. B. Mann) New York: Wiley, 1968, pp. 15-46.
- [6] E. R. Berlekamp, *Algebraic Coding Theory*. New York: McGraw-Hill, 1968.
- [7] P. Delsarte, J. M. Goethals, and F. J. MacWilliams, "On generalized Reed-Muller codes and their relatives," *Information and Control*, vol. 16, pp. 403-442, July 1970.
- [8] H. F. Harmuth, "Applications of Walsh functions in communications," *IEEE Spectrum*, vol. 6, pp. 82-91, November 1969.
- [9] K. W. Henderson, "Comment on computation of the fast Walsh-Fourier transform," *IEEE Trans. Computers*, vol. C-19, p. 850, September 1970.

A Method for Decoding of Generalized Goppa Codes

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Abstract—It is shown how the theory of continued fractions for polynomials can be used for the decoding of a class of codes which contains Goppa codes.

I. INTRODUCTION

It has been shown [1] that Goppa codes are a subclass of the codes which are generated by the Chinese remainder theorem [2]. These latter codes have the same relationship to Goppa codes as the codes constructed by Stone [3] do to Reed-Solomon codes. A method of decoding these generalized Goppa codes which uses the theory of continued fractions for polynomials is presented